Hickathon #6 : Mental health & well-being

How can we come up with innovative solutions to support students ?

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Groupe 42

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Table des matières

1 First task - Regression Task - Prediction of MathScore

1.1 Introduction

PISA assesses students' ability to apply knowledge in real-life contexts, considering external (socioeconomic, cultural, school, national system) and internal factors (motivation, self-esteem, learning strategies). Using a merged table from 2015, 2018, and 2022 (1.7M students, 250+ features), our objective is to predict each student's **MathScore** while extracting interpretable, socially impactful insights to support educational decisions.

1.2 Exploratory Data Analysis (EDA)

During our first iteration on the dataset, we were surprised by the very high proportion of missing values. Rather than treating them as rows to discard, we hypothesised that missingness itself might carry behavioural information.

By examining the structure of NaNs more closely, we identified two important patterns :

— When a student stops answering a math/reading/science test, all the following questions columns also become NaN. We therefore created a new feature to encode the exact point where answering stops and reduce sparsity.

— Several question blocks appear only in certain years. As a result, many columns are entirely empty when looking at a special year. Once we segmented the data by year, these columns could safely be removed for modelling and analysis, which greatly reduced the amount of NaN. This pushed us later on to train a dedicated model for each year.

During EDA, we also identified additional challenges that would influence our modelling choices :

— The MathScore target contains a large number of zeros. We chose not to split the task into classification + regression during this short hackathon, but still considered it a direction worth evaluating later.

— The **CNT** (country) variable probably contains a strong contextual signal but has very high cardinality. One-hot encoding all countries would have produced extremely large sparse vectors and high risk of overfitting, so we explored more compact representations instead.

— Even after removing year-specific empty columns, the number of remaining features was still large, meaning some form of feature selection would likely be necessary.

1.3 Modelling and choices

For modelling, we selected **LightGBM** as our main regressor as it scales efficiently to millions of samples and it handles missing values natively.

Our core contribution is that we used a two-stage approach : a global regressor trained on the full dataset captures the main explainable signal, then we created residual models per year, as we saw that features we not the same for each year, to capture insights inside each subgroup (2015,2018,2022). It also allow us to drop all the NaN columns per year and enhance model performance.

1.3.1 Feature engineering and selection

To reduce dimensionality without losing too much signal, we applied the following strategy :

— We computed Spearman correlations as they are non linear compared to Pearson to identify features that had no monotonic relationship with MathScore.

— We removed features that were nearly duplicates of others, based on pairwise correlations, to avoid redundancy and unstable importance measures.

— For country, instead of a high-dimensional encoding, we generated a numeric rank feature summarising relative country performance for each training fold. This gave the model access to country-level patterns while keeping the representation compact and less leak-prone.

1.3.2 Hyperparameter optimisation

Initially, we trained LightGBM using near-default parameters to evaluate feasibility. The model reached $\mathrm{R}^2 \approx 0.77$ immediately. We then applied **Optuna tuning** to adjust depth, learning rate, and regularisation parameters under reasonable compute constraints. This tuning provided almost no improvement. This reinforced our suspicion that the model was already capturing most of the explainable signal very early, and that the remaining variance might be mostly noise or require very different modelling assumptions.

1.3.3 Expert Aggregation

Still, in order to reach slightly better results, we use an expert aggregation of 10 LightGBM fine tuned and chosen via Optuna. This allows us to gain 1% performance and to reach a $\mathrm{R}^2 \approx 0.78$ while still preventing overfitting.

1.4 Results and interpretability

Our final model eventually achieved an **R^2 of approximately 0.78 on test data**, which is aligned with the scores observed across other teams and architectures. The fact that all models quickly converged to the same range suggests some underlying explanations : the predictable signal might be intrinsically limited or a small number of features probably carry most of the modelling power.

To validate this, we analysed both SHAP values and tree feature importance.

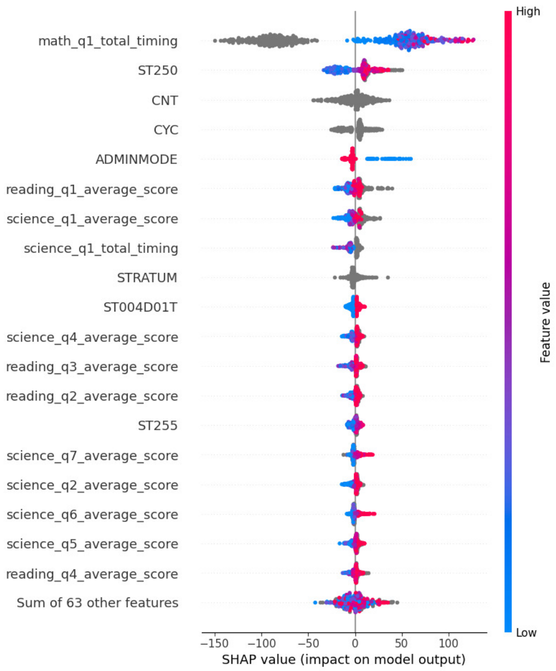

FIGURE 1 – SHAP Values of the most predictive features

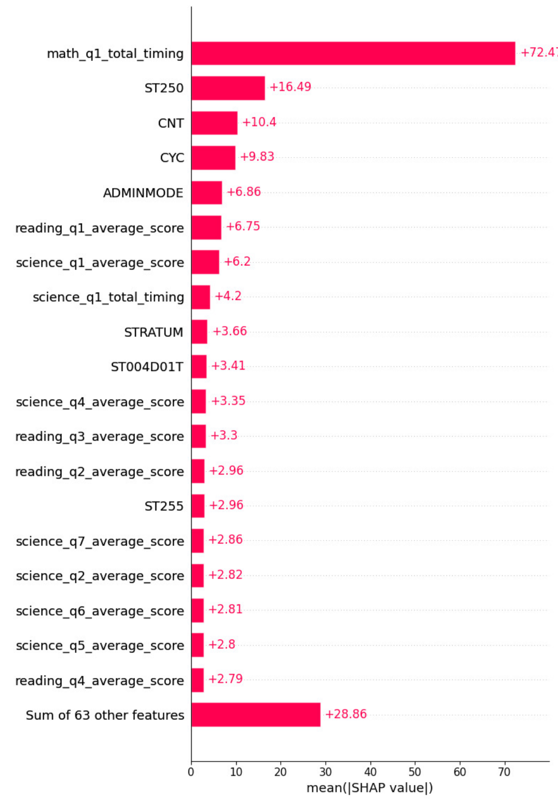

FIGURE 2 – Predictive power of the features

This feature analysis validates our main assumptions from EDA. The variable `math_q1_total_timing`, intentionally kept in the test set, stands out as the strongest predictor. We also confirm the importance of country and cycle features (`CNT`, `CYC`), showing that the model captures cross-system performance patterns despite their initial categorical form. Finally, `ST250`, which represents educational resources available at home, carries strong predictive power and follows an intuitive positive trend consistent with SHAP values. The `ST255` feature, which represents the number of books available at home, plays also a top 10 rôle in the model prediction. Additional sanity checks are provided in Figure 1.

1.5 Conclusion

All in all, we managed to compute a viable, socially meaningful regression approach that leverages year-aware pruning to maximise signal capture while delivering explainable insights for education policy.

2 CARDS GAME : HARMONIA

A cooperative card game to enhance collective well-being

Overview of the problems students could face thanks to the PISA dataset

Code	Description	Catégorie
WB153	Satisfaction with overall appearance, body, weight, and clothing fit.	Image de soi / Estime personnelle
WB154	Frequency of headaches, stomach pain, nervousness, depression, irritability, sleep issues, dizziness, anxiety.	Symptômes psychologiques / Stress
WB155	Satisfaction with life aspects : health, relationships, education, possessions, school experience.	Bien-être global
ST345	How students experience and manage stress, panic, nervousness, and recovery after pressure.	Gestion du stress / Régulation émotionnelle
WB178	Evaluation of well-being during the previous day (accomplishment, energy, satisfaction).	Bien-être quotidien
WB166	Emotions experienced during the last mathematics class.	Émotions en contexte scolaire
WB168	Emotions experienced during the last test language class.	Émotions en contexte scolaire
ST038	Experiences or witnessing of bullying, exclusion, aggression, threats, physical harm.	Harcèlement / Sécurité
ST034	Perceived sense of belonging and social connection at school.	Appartenance sociale / Climat scolaire
ST266	Exposure to school violence (vandalism, fights, gangs, threats, weapons).	Violence scolaire / Climat scolaire
WB160	Frequency and mode of communication with friends.	Relations sociales / Sociabilité
ST336	Extent to which friends and family encourage creativity, imagination, new ideas.	Soutien social à la créativité
ST267	Teachers' respect, friendliness, and concern for student well-being.	Relation enseignant-élève / Bienveillance
ST211	Teacher making students feel confident, listened to, and understood.	Soutien émotionnel des enseignants
ST270	Teacher actively supporting and ensuring understanding of lessons.	Support pédagogique / Aide académique
ST100	Individualized attention and support from teacher during lessons.	Support pédagogique / Aide personnalisée
WB164	Frequency of experiencing financial anxiety for the family.	Anxiété financière / Stress familial
WB163	Extent of parental support for independence versus controlling behavior.	Style parental / Autonomie émotionnelle
WB162	Ease of talking about personal problems with family, friends, teachers.	Communication / Recherche d'aide
ST353	Family support for learning during COVID (homework help, scheduling, supervision).	Soutien familial / Apprentissage
ST322	Impact of digital device habits on focus and well-being in daily life.	Effets du numérique / Hygiène digitale
IC177	Total daily time spent on digital activities (gaming, social media, browsing, learning, content creation).	Usage numérique / Temps d'écran
ST158	Skills taught in school for responsible internet use (search strategies, safety, source evaluation).	Compétences numériques / Citoyenneté digitale
ST311	Ability to understand others' feelings, perspectives, emotional needs (empathy).	Compétences socio-émotionnelles / Empathie
ST343	Willingness to help others, cooperate, work in teams, avoid conflict or selfish behavior.	Coopération et comportement social
ST188	Self-confidence and ability to overcome challenges, persistence, resilience.	Auto-efficacité / Résilience
ST183	Extent to which failure triggers fear of judgment, self-doubt, and anxiety about the future.	Peur de l'échec / Anxiété de performance

TABLE 1 – Liste des variables de la matrice de corrélation avec descriptions et catégories

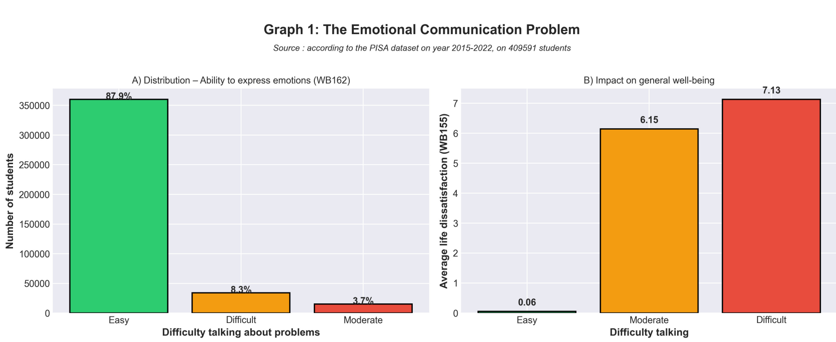

FIGURE 3 – The need of talking against dissatisfaction in every day life

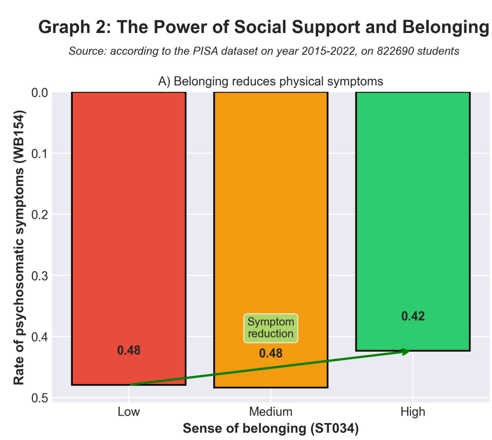

FIGURE 4 – The effect of belonging in reducing psychomatic symptoms

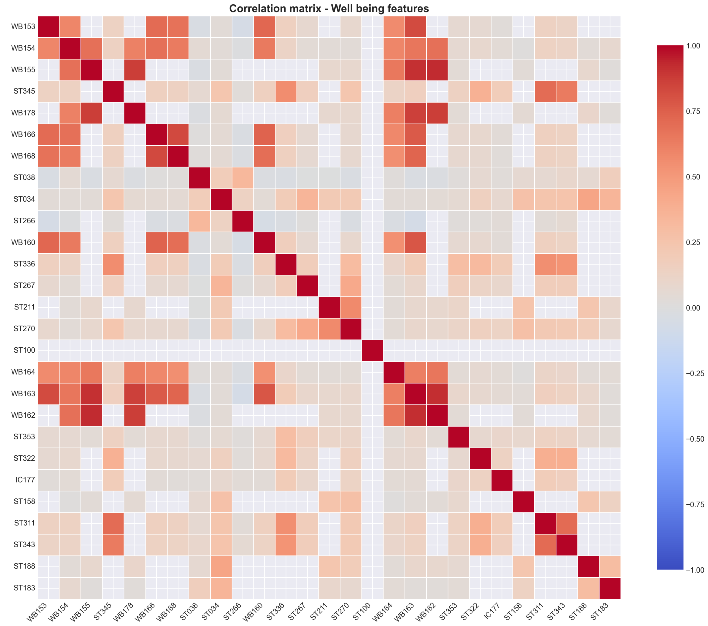

FIGURE 5 – Correlation matrix on selected well being features (Cf. glossaire)

The figure 3 shows the impact of communication on the well being of students : as the talking difficulty rises ; the dissatisfaction of one's explodes. It shows a hundredfold increasing as the ease of talking jumps from easy to difficult. One of our goal is then helping students talking about their feelings.

The figure 4 explains how the sentiment of belonging affects the advent of psychosomatic symptoms of a student. As the sentiment of belonging is indeed high, less one would experience psychosomatic symptomps. The graph shows as decrease of 6 percentage points of the symptoms as they feel included. Thus our next goal is helping students feel included with their peers.

Figure 5 is a correlation matrix showing linear correlation between all our selected well being features. It helped us determined what problems students are facing and so how we could concretely help them.

Game summary

Harmonia is a cooperative card game where players take on the role of **Guardians of Well-Being**. Their mission : protect their collective **Inner Garden** from **Storms** (stress, isolation, bullying...) by using **Zen Action** cards and emotional **Shields**. By nurturing the **Flower of Serenity** together, they develop practical skills in resilience, cooperation, and emotional intelligence. A game designed for school settings, grounded in psychometric concepts, and above all... **fun and engaging** !

SOLUTION JUSTIFICATION : A SOCIAL GAMIFICATION APPROACH WITH A CARD GAME

Student well-being has become a growing priority in schools : academic stress, performance anxiety, social isolation, cyberbullying, and digital overload can directly impact motivation, performance, and self-esteem [1]. Despite new approaches emerging—digital support platforms, self-assessment tools, or discussion forums—many limitations remain. These solutions often feel individual and screen-based, which can lack personal connection and sometimes even reinforce isolation, reducing the impact of peer and social support benefits (cf. Figure 4).

Harmonia addresses this gap by offering a socially innovative, tangible solution built around **collective interaction and positive gamification**. The cooperative card format turns complex psychological concepts—such as resilience, emotional regulation, and empathy—into simple shared actions and team-based mechanics. From a project and operational perspective, this brings several strategic advantages :

1. UNICEF & WHO, 2021, *Adolescent Mental Health : Global Status Report.*

1. **Strengthening peer support and belonging :** By encouraging cooperation and emotional expression, *Harmonia* directly supports group cohesion and the sense of belonging—key levers for preventing isolation. [2].
2. **A non-stigmatizing educational approach :** The game format allows teams to address sensitive mental health and social issues in a safe, engaging environment, boosting voluntary participation and natural peer-to-peer learning of socio-emotional skills.
3. **Market alignment and potential value :** The educational serious game market is growing rapidly, estimated at more than 25 billion USD by 2025, while physical and social board game formats follow a sustained positive adoption trend, showing demand for interactive learning tools and collaborative solutions [3].
4. **Turn-key deployment and adaptability :** The product is designed for school operations (middle schools, high schools) and school psychologists, as a "ready-to-run" support toolkit that can integrate immediately into classrooms, workshops, or well-being sessions, with potential adaptations to local educational well-being programmes.

In summary, *Harmonia* positions itself as an original product at the intersection of educational support, mental health, and social interaction : it transforms prevention and socio-emotional support into a collective, physical, and engaging experience—while unlocking justified, compact, and scalable deployment opportunities for real-world educational operations [4].

• Game Rules : Harmonia

Game Components

- **1 "Inner Garden" Board** : Represents the group's well-being through the **Serenity Path** (scaled from 0 to 20).
- **1 "Flower of Serenity" Token** : Tracks collective progress on the board.
- **5 "Balance" Cards** : Individual boards showing each player's mental energy gauge (starting at 3 HP).
- **90 Game Cards** split into three decks :
  - **30 "Storm" Cards** (Threats) : Academic stress, bullying, fatigue, pressure, etc.
  - **30 "Shield" Cards** (Resources) : Protection, resilience, peer support.
  - **30 "Zen Action" Cards** (Solutions) : Breathing exercises, communication, creativity, etc.

Quick Rules

Game Objective

✔ **Win :** Cooperate to make the **Flower of Serenity bloom** by raising it from 10 to **20 Serenity**.

✘ **Lose :** The Flower falls to **0 Serenity or** all players reach 0 on their well-being gauge.

Turn Structure (3 Phases)

1. Draw Phase (Preparation)
Each player draws to complete a hand of **3 cards**. This is also a moment for quick strategy sharing ("I can manage stress !").

2. Storm Phase (The Event)
The active player reveals the top card of the **Storm deck**.
— *Immediate effect :* Apply the listed penalties (e.g., loss of mental energy, enforced silence, etc.).
— *Challenge :* The card defines a **Threat Value** (e.g., 4) that the group must match or exceed to dispel it.

3. Response Phase (Collective Action)
To face the Storm, each player must choose **only 1 action** among the 4 possible ones.

2. OECD, 2021, *Students' Sense of Belonging and School Climate.*
3. Newzoo, 2023, *Global Games Market Report* ; Statista, 2023, *Board Games Industry Growth.*
4. UNICEF & WHO, 2021 ; OECD, 2022 ; Newzoo, 2023.

The 4 Possible Actions

① Play a Zen Action	Place a *Zen Action* card in front of you. — We sum the values of all Action cards played by the group. — **If Total ≥ Threat** : The Storm is defeated and Serenity increases (+1). — **If Total < Threat** : The Storm remains active and will strike again next turn.
② Activate a Shield	Place a *Shield* card to cancel a specific penalty effect (e.g., prevent loss of mental energy) or prepare future defence.
③ Support a Friend	Discard one card from your hand to immediately restore **+1 Balance** to a struggling player. *"I am here for you."*
④ Meditate (Rest)	Play nothing. You pass your turn to restore **+1 Balance** to your own mental gauge.

• Card Gallery

Here is a curated selection of the game cards with a unified, harmonised design. They are inspired from the PISA dataset.

The Storms (danger)

Fear of failure
Codes : WB155 / ST188

Effect : Every player loses **1 Balance**.
Threat : If one player is at 1 well-being, additionnal loss of **-2 Serenity**.

Hyper-connection
Codes : ST270 / WB154

Effect : "Zen action" cards cost **2 cards** instead of 1 this turn.
Cognitive surcharge and loss of concentration.

Inner Solitude
Codes : ST034 / WB162

Effect : **No exchange** of cards is authorised this turn.
Every player lose 1 Balance.

Chronical Fatigue
Codes : WB154 / ST345

Effect : Every player starts at **-1 Balance**.
If serenity <10, add -1 well-being.

Scholar Pressure
Codes : ST270 / WB155

Effect : **-2 Global Serenity**.
If two players are below 3 Balance, add -2 Serenity.

Autonomy conflict
Codes : WB163

Effect : Players can only play cards with a value of 1 this turn.
Feeling of excessive parental control.

The Shields (defense)

Positive affirmation
Type : Permanent

Effect : At the start of each turn, choose a player. They gain **+1 Balance**.

Group solidarity
Type : Permanent

Effect : If a Storm affects multiple players, reduce the global penalty by half.

Disconnection
Type : One usage

Effect : Once per game, **completely cancels** the effect of a Storm related to cognitive overload.

Creative Cohesion
Type : Permanent

When two players play a Zen Action on the same turn, add **+1 Serenity**.

Ally Professor
Type : Usable / **Codes :** ST267

Effect : Cancels an "Academic Pressure" type Storm thanks to teacher benevolence.

Body spirit
Type : Single use / **Codes :** ST343

Effect : All players can play 2 cards this turn instead of one.

Zen Actions (Answers)

Coherent Breathing
Duration : 2 min

Effect : **+2 Collective serenity**.
"Inhale for 5, exhale for 5."

Sincere Compliment
Type : Social

Effect : Choose a player. They gain **+2 Balance** and remove one active penalty.

Digital Break
Type : Mental

Effect : The next Cognitive Storm has its effects reduced by **50%**.

Emotional Debrief

Type : Group

Effect : If two players are playing, +1 Serenity. If 3+, +3 **Serenity**.

Stretching

Type : Physic

Effect : All players get **+1 Balance**.

Moment of Gratitude

Codes : WB178

Effect : **+1 Serenity** for each player with their well-being gauge full.

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Business roadmap

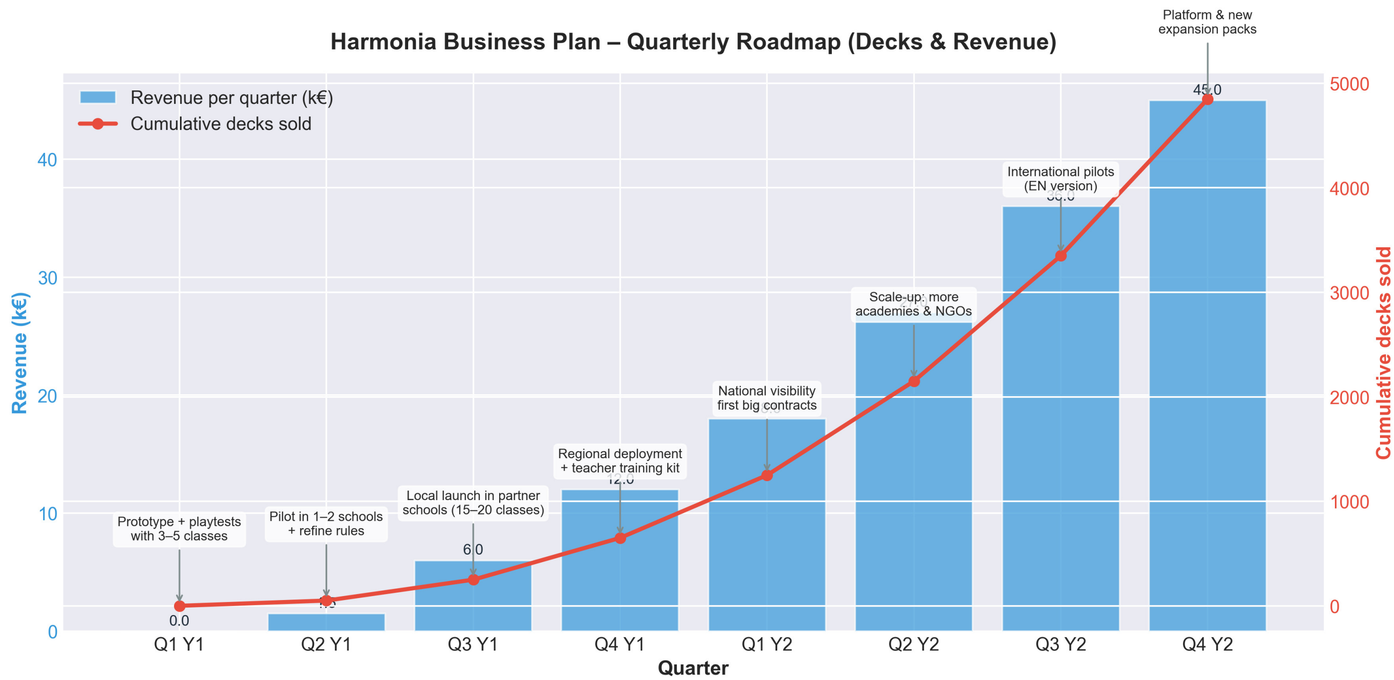

FIGURE 6 – Business roadmap

Figure 6 is the roadmap we may pretend for our product : Harmonia. We can expect lauching a prototype of our product on a few sample of classes (3-5 classes). Then use the feedback from those classes to upscale little by little our product accordingly to the map.